

Code: 1208

NOV-DEC 2024 EXAMINATION
I B.E /B. TECH EXAM
MA 10511:-MATHEMATICS-II

Time: 3 Hrs.]

[Max. Marks: 70]

TOTAL NO. OF QUESTIONS IN THIS PAPER:5

Note: Attempt all questions. All questions carry equal marks. Each question carries five subparts a, b, c, d and e. Parts a, b and c are compulsory and attempt any one from d and e.

S. No.	Questions	Mark	CO	BL	PI
Q.1	(a) Define Hermitian and Skew Hermitian Matrix.	(02)	1	1	1.1.1
	(b) Examine for linear dependence or independence of vectors: $x_1 = (1,2,3), x_2 = (2,-2,6)$ If dependent, find the relation between them.	(02)	1	1,2	1.1.1
	(c) Find the Eigen values and Eigen vector of matrix $A = \frac{1}{24} \begin{bmatrix} 2 & 4 \\ -2 & 8 \end{bmatrix}$	(03)	1	1,2	1.1.1
	(d) Reduce the following matrix into its normal form and hence find its Rank. $A = \begin{bmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}$	(07)	1	3	1.1.1
	OR				
	(e) Determine the values of λ and μ such that the system $2x - 5y + 2z = 8, 2x + 4y + 6z = 5, x + 2y + \lambda z = \mu$ have (i) no solution (ii) a unique solution (iii) infinite number of solutions.	(07)	1	4	1.1.2
Q.2	(a) Define Diagonalization of Matrix.	(02)	2	1	1.1.1
	(b) If $A = \begin{bmatrix} 2 & 4 \\ -3 & 2 \end{bmatrix}$, find A^{-1} .	(02)	2	1,2	1.1.1
	(c) Verify Cayley Hamilton theorem for the matrix $A = \begin{bmatrix} 3 & -1 \\ -1 & 5 \end{bmatrix}$	(03)	2	2	1.1.1
	(d) Let $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$. Find matrix P such that $P^{-1}AP$ is diagonal matrix.	(07)	2	1,3	1.1.1
	OR				
	(e) Reduce the following quadratic form to canonical form by orthogonal transformation. Also, find the rank, index, signature, and value class (nature) of the quadratic form. $x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2 - 2x_1x_3 + 2x_2x_3$	(07)	2	1,3	1.1.2

Q.3	(a)	Obtain the differential equation associated with the primitive $y = A \cos x + B \sin x$	(02)	3	1	1.1.1
	(b)	Solve by exactness $(5x^4 + 3x^2y^2 - 2xy^3)dx + (2x^3y - 3x^2y^2 - 5y^4)dy = 0$	(02)	3	2	1.1.1
	(c)	Solve $(x^2 - y \cdot x^2)dy + (y^2 + xy^2)dx = 0$	(03)	3	1.2	1.1.1
	(d)	Solve $\frac{dy}{dx} = 2y \tan x + y^2 \tan^2 x$	(07)	3	3	1.1.1
OR						
	(e)	Solve $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 3y = e^{2x} + \cos x + x^2$	(07)	3	1.2	1.1.2
Q.4	(a)	Write down the basic steps for solving Legendre's homogeneous linear differential equation.	(02)	4	1	1.1.1
	(b)	Solve the simultaneous equation $\frac{dx}{dt} + 10y = 0, \frac{dy}{dt} - 10x = 0$.	(02)	4	1	1.1.1
	(c)	Find the Complimentary function of $x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} - 12y = 0$	(03)	4	2	1.1.1
	(d)	The equation of electromotive force in terms of current i for an electrical circuit having resistance R and a condenser of capacity C , in series is $E = Ri + \int \frac{i}{C} dt$. Find the current i at any time t , when $E = E_0 \sin wt$.	(07)	4	3	1.1.2
OR						
	(e)	Apply the method of variation of parameters $\frac{d^2y}{dx^2} + y = (x - \cot x)$	(07)	4	3	1.1.1
Q.5	(a)	An urn contains 10 black and 10 white balls. Find the probability of drawing two balls of the same colour.	(02)	5	2	1.1.1
	(b)	A die is tossed. If the number is odd, what is the probability that it is prime?	(02)	5	2	1.1.1
	(c)	In a normal distribution, 31 % of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. [Given that $Z(0.19) = 0.496, Z(0.42) = 1.405$]	(03)	5	3	1.1.1
	(d)	A can hit a target 3 times in 5 shots, B 2 times in 5 shots and C three times in 4 shots. All of them fire one shot each simultaneously at the target. What is the probability that (i) 2 shots hit (ii) At least two shots hit?	(07)	5	3	1.1.1
OR						
	(e)	(i) Derive the formula of Mean of Binomial Distribution. (ii) If the mean and variance of a binomial distribution are 4 and 2 respectively, find the probability of (i) exactly 2 successes (ii) less than 2 successes (iii) at least 2 successes.	(03+04)	5	2,3	1.1.2

NOV-DEC 2024 EXAMINATION
I B.E /B. TECH EXAM
MA 10501:-MATHEMATICS-II

Time: 3 Hrs.]

[Max. Marks: 70]

TOTAL NO. OF QUESTIONS IN THIS PAPER:5

Note: Attempt all questions. All questions carry equal marks. Each question carries five subparts a, b, c, d and e. Parts a, b and c are compulsory and attempt any one from d and e.

S. No.	Questions	Marks	CO	BL	PI
Q.1 (a)	Define Symmetric Matrix.	(02)	1	1	1.1.1
(b)	Find the inverse of $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$.	(02)	1	1,2	1.1.1
(c)	Find the Eigen Values of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$.	(03)	1	1,2	1.1.1
(d)	Find the Rank of a Matrix $A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & -2 \\ 2 & 4 & -3 \end{bmatrix}$.	(07)	1	3	1.1.2
OR					
(e)	State and Prove Cayley Hamilton theorem.	(07)	1	4	1.1.1
Q.2 (a)	Write Order and Degree of the following equation	(02)	2	1	1.1.1
	$\frac{d^2y}{dx^2} + a^2x = 0$				
(b)	Form the differential equation by eliminating arbitrary constant. $y = Ax + A^2$	(02)	2	1,2	1.1.1
(c)	Solve the differential equation $(2xy - 3x^2)dx + (x^2 - 2y)dy = 0$	(03)	2	2	1.1.1
(d)	Solve $\frac{dy}{dx} + \frac{1}{x}y = x^3 - 3$	(07)	2	1,3	1.1.1
OR					
(e)	Solve $\frac{d^2y}{dx^2} - 4y = e^x + \sin 2x$	(07)	2	1,3	1.1.2
Q.3 (a)	Write L-R Series Circuit Equation.	(02)	3	1	1.1.1
(b)	Solve the following differential equation $\frac{dx}{dt} = y + 1, \frac{dy}{dt} = x + 1$	(02)	3	2	1.1.2
(c)	Write down the basic steps for solving method of variation of parameters.	(03)	3	1,2	1.1.1
(d)	Solve $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4$	(07)	3	3	1.1.2
OR					
(e)	An inductance of 2 henries and a resistance of 20 ohms are connected in series with an e.m.f E volts. If the current is zero when $t = 0$, find the current at the end of 0.01 sec. if $E = 100$ Volts.	(07)	3	1,2	1.1.2

- Q.4 (a) Write down the formula of Poisson Distribution.
 (b) Find the probability of getting sum of 9, when two dice are rolled.
 (c) Write down any three properties of Normal distribution.
 (d) Derive the formula of mean and variance of binomial distribution.

(02) 4 1 1.1.1
 (02) 4 1 1.1.1
 (03) 4 2 1.1.1
 (07) 4 3 1.1.1

OR

- (e) Find the straight line $y = a + bx$ of the following data.

x	1	2	3	4	5
y	14	27	40	55	68

(07) 4 3 1.1.2

- Q.5 (a) Subtract $Z_1 = \frac{3}{2} - \frac{4}{3}i$ from $Z_2 = \frac{-4}{3} + \frac{10}{4}i$
 (b) State De-Moivre's theorem.
 (c) Multiply $3 + 4i$ by $7 - 3i$
 (d) Separate $\tan^{-1}(a + ib)$ into real and imaginary parts.

(02) 5 5 1.1.1
 (02) 5 2 1.1.1
 (03) 5 3 1.1.1
 (07) 5 3 1.1.1

OR

- (e) Sum the series

$$\sin \alpha + x \sin(\alpha + \beta) + \frac{x^2}{2!} \sin(\alpha + 2\beta) + \dots \infty$$

(07) 5 3 1.1.2

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APRIL 2024 EXAMINATION
B.E/B.TECH EXAMINATION
MA 10501 Mathematics-II

Time : 3 hour

Max. Marks: 70

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

Note: Attempt all questions. All questions carry equal marks. Each question carries five subparts a, b, c, d and e. Parts a, b and c are compulsory and attempt any one from d and e.

	Questions	Marks	CO	BL
Q.1(a)	Define Orthogonal and Unitary Matrix with example.	02	CO1	1
(b)	Show that the matrix $B^{\theta}AB$ is Hermitian according as A is Hermitian matrix.	02	CO1	2
(c)	Show that the equation $2x + 6y = -11$, $6x + 20y - 6z = -3$, $6y - 18z = -1$ are not consistent.	03	CO1	2
(d)	Reduce the matrix to normal form and find its rank $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 3 & 4 & 0 & -1 \\ -1 & 0 & -2 & 7 \end{bmatrix}$	07	CO1	3
	OR			
(e)	Find the Eigen values and Eigen vectors of the matrix $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$	07	CO1	3
Q.2 (a)	Solve: $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$	02	CO2	2
(b)	Find the integrating factor(I.F) of the linear differential equation $\frac{dy}{dx} + y \tan x = \sin 2x$.	02	CO2	2
(c)	Solve the Exact differential equation $(\cos x + y \sin x)dx = (\cos x)dy$	03	CO2	2
(d)	Solve: $\frac{dy}{dx} - \frac{y \tan y}{1+x} = (1+x)e^x \sec y$	07	CO2	3
	OR			
(e)	Solve: $(D^2 - 4D + 4)y = x^2 + e^x + \cos 2x$	07	CO2	3
Q.3 (a)	Find the Complimentary Function (C.F) of $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^2 + 2 \log x$	02	CO3	2
(b)	Find the particular integral(P.I) of $(1+x)^2 \frac{d^2y}{dx^2} + (1+x) \frac{dy}{dx} + y = \sin[2\{\log(1+x)\}]$.	02	CO3	2
(c)	Apply the method of variation of parameter. Find the value of u $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$	03	CO3	2

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(d) Solve: $\frac{dx}{dy} + y = \sin t$, $\frac{dy}{dx} + x = \cos t$

07 CO3 3

(e) A condenser of capacity C farads with V_0 is discharged through a resistance R ohms. Show that if q coulomb is the charge on the condenser, i amperes the current and V the voltage at time t.
 $q = CV, V = Ri$ and $i = \frac{dq}{dt}$, hence show that $V = V_0 e^{-t/RC}$

07 CO3 3

Q.4(a) A die is tossed thrice. A success is getting 1 or 6 on a toss. Find the mean.

02 CO4 2

(b) Write the recurrence formula of Binomial distribution.

02 CO4 1

(c) In a sample of 1000 cases, the mean of certain test is 14 and standard deviation is 2.5. Assuming the distribution to be normal. Find how many score above 18? (Area right to z_1 is 0.4452)

03 CO4 2

(d) Fit a Poisson distribution in the set of observation

x	0	1	2	3	4
f(x)	122	60	15	2	1

07 CO4 2

(e) (i) Write the normal equation to fit up the parabola
 (ii) Using least square method to fit a straight line in the following data.

07 CO4 1,3

x	2	7	9	1	5	12
f(x)	13	21	23	14	15	21

Q.5(a) Separate into real and imaginary part $\cos(x + iy)$

02 CO5 2

(b) If $\tan(A + iB) = x + iy$, prove that

$$\tan 2A = \frac{2x}{1 - x^2 - y^2}$$

02 CO5 2

(c) Prove that $\cosh^{-1} \sqrt{1 + x^2} = \sinh^{-1} x$

03 CO5 2

(d) If ω is a cube root of unity, prove that $(1 - \omega)^6 = -27$

07 CO5 3

OR

(e) (i) If $x = 2 \sin \alpha \cosh \beta$, $y = 2 \cosh \alpha \sinh \beta$, show that

$$\operatorname{cosec}(\alpha - i\beta) + \operatorname{cosec}(\alpha + i\beta) = \frac{4x}{x^2 + y^2}$$

07 CO5 3

(ii) Sum the series $\frac{\sin \theta}{1!} - \frac{\sin 2\theta}{2!} + \frac{\sin 3\theta}{3!} - \frac{\sin 4\theta}{4!} + \dots \dots \dots \infty$

Code: 558

NOV-DEC 2023 EXAMINATION
I B.E./B. TECH EXAM
MA 10501:-MATHEMATICS-II

Time: 3 Hrs.]

[Max. Marks: 70]

TOTAL NO. OF QUESTIONS IN THIS PAPER:5

Note: Attempt all questions. All questions carry equal marks. Each question carries five subparts a, b, c, d and e. Parts a, b and c are compulsory and attempt any one from d and e.

S. No.	Questions	Mark	CO	BL
Q.1	(a) Find the Eigen values of the matrix $\begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$	(02)	1	1,2
	(b) Prove that $A = \begin{bmatrix} 4 & 1-3i \\ 1+3i & 7 \end{bmatrix}$ is Hermitian matrix	(02)	1	1,2
	(c) Show that the vectors $x = (2, 3, 1, -1)$, $y = (2, 3, 1, -2)$, $z = (4, 6, 2, 1)$ are linearly dependent. Express one of the vectors as linear combination of the others.	(03)	1	2
	(d) Find Non-Singular matrices P and Q such that PAQ is normal form, Hence find the Rank of A where $A = \begin{bmatrix} 2 & -2 & 3 \\ 3 & -1 & 2 \\ 1 & 2 & -1 \end{bmatrix}$	(07)	1	3
	OR			
	(e) (i) State Cayley Hamilton theorem. (ii) Investigate the values of λ and μ so that the equations $2x + 3y + 5z = 9$, $7x + 3y - 2z = 8$, $2x + 3y + \lambda z = \mu$ have (i) No solution (ii) a Unique solution (iii) an infinite number of solution.	(07)	1	1,4
Q.2	(a) Write Order and Degree of the following equation $x^2 \left(\frac{d^2 y}{dx^2} \right)^2 + y \left(\frac{dy}{dx} \right)^4 + y^4 = 0$	(02)	2	1
	(b) Form the differential equation by Eliminating arbitrary constant. $y^2 = Ax^2 + Bx + C$	(02)	2	1,2
	(c) Solve the differential equation $(x^2 - yx^2)dy + (y^2 + xy^2)dx = 0$	(03)	2	2
	(d) Solve $\frac{dy}{dx} + \frac{1}{x} \tan y = \frac{1}{x^2} \tan y \sin y$	(07)	2	1,3
	OR			
	(e) Solve $\frac{d^2 y}{dx^2} - 4 \frac{dy}{dx} + 4y = e^{2x} + x^3 + \cos 2x$	(07)	2	1,3
Q.3	(a) Find the Complimentary Function of $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$.	(02)	3	1
	(b) Solve the following differential equation $\frac{dx}{dt} = y + 1$, $\frac{dy}{dt} = x + 1$	(02)	3	2

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Q.4	(c)	Form a differential equation of an R-C circuit having voltage across resistor (R) and capacitor(C) are given by $V_R = R_i$ and $V_C = \frac{1}{C} \int i dt$ respectively (where i denotes the current).	(03)	3	1,2	1.1.1
	(d)	Solve $(3x+2)^2 \frac{d^2y}{dx^2} + 3(3x+2) \frac{dy}{dx} - 36y = 3x^2 + 4x + 1$.	(07)	3	3	1.1.1
	OR					
	(e)	Apply the method of variation of parameter to solve $\frac{d^2y}{dx^2} - 3 \frac{dy}{dx} + 2y = \frac{e^x}{1+e^x}$	(07)	3	1,2	1.1.1
	(a)	Write any three properties of Normal curve.	(02)	4	1	1.1.1
	(b)	Students of a class were given an aptitude test. Their marks were found to be normally distributed with mean 60 and standard deviation 5. What percentage of students scored more than 60 marks?	(02)	4	1	1.1.1
	(c)	Assume that the probability of an individual coal miner being killed in a mine accident during a year is $\frac{1}{2400}$. Use appropriate statistical distribution to calculate the probability that in a mine employing 200 miners, there will be at least one fatal accident in a year. ($e^{-\frac{1}{12}} = 0.92$)	(03)	4	2	1.1.1
	(d)	Derive the formula of mean and variance of binomial distribution.	(07)	4	3	1.1.1
	OR					
	(e)	(i) Write down the recurrence formula of Poisson distribution. (ii) By the method of least square, find the straight line that best fits the following data.	(07)	4	3	1.1.1

x	1	2	3	4	5
y	14	27	40	55	68

Q.5	(a)	Express $\frac{(3+4i)(2+i)}{(1+i)}$ in the form of $a+ib$.	(02)	5	2	1.1.1
	(b)	Find the complex conjugate of $\frac{2+3i}{1-i}$.	(02)	5	2	1.1.1
	(c)	Expand $\tan 5\theta$ in powers of $\tan \theta$ using D-Moivre's theorem.	(03)	5	3	1.1.1
	(d)	If $\alpha, \alpha^2, \alpha^3, \alpha^4$ are the roots of $x^5 - 1 = 0$ find them and show that $(1-\alpha)(1-\alpha^2)(1-\alpha^3)(1-\alpha^4) = 5$.	(07)	5	3	1.1.1
	OR					
	(e)	(i) Separate $\sin(x+iy)$ into real and imaginary parts. (ii) Sum the series	(07)	5	3	1.1.1

$$\sin \alpha - \frac{\sin(\alpha+2\beta)}{2!} + \frac{\sin(\alpha+4\beta)}{4!} - \dots \infty$$

JUNE-JULY 2023 EXAMINATION
I B. TECH EXAM
MA 10501:-MATHEMATICS-II

[Max. Marks: 70]

Time: 3 Hrs.]

TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

Note: Attempt all questions. All questions carry equal marks. Each question carries five subparts a, b, c, d and e. Parts a, b and c are compulsory and attempt any one from d and e.

S. No.	Questions	Mark	CO	BL	
		(02)	1	1	1.
Q.1	(a) Define Symmetric and Skew-Symmetric matrix with example.	(02)	1	1	1.
	(b) Find the Eigen value of the Matrix $\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$	(03)	1	2	1.
	(c) Find the Rank of Matrix $A = \begin{bmatrix} 1 & 5 & 4 \\ 0 & 3 & 2 \\ 2 & 13 & 10 \end{bmatrix}$	(07)	1	1,2	1.
	(d) State Cayley Hamilton Theorem and Verify Cayley Hamilton theorem for the Matrix $A = \begin{bmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{bmatrix}$ and Hence find A^{-1} .				
	OR				
	(e) Find for what values of λ and μ the system of linear equations $\begin{aligned} x + y + z &= 6 \\ x + 2y + 5z &= 10 \\ 2x + 3y + \lambda z &= \mu \end{aligned}$ Has (i) a Unique solution (ii) No solution (iii) Infinite solution. And also find the solution for $\lambda = 2$ and $\mu = 8$.	(07)	1	3	1.
Q.2	(a) Define Linear differential equation of first order and first degree with example.	(02)	2	1	1.
	(b) Form the differential equation by eliminating arbitrary constants, $y = A \cos x + B \sin x$	(02)	2	2	1.
	(c) By exactness, Solve the differential equation $(x + y - 2)dx + (x - 2y - 3)dy = 0.$	(03)	2	2	1.
	(d) Solve $\frac{dy}{dx} + x \sin 2y = x^3 \cos^2 y$	(07)	2	3	1.
	OR				
	(e) Solve $\frac{d^2y}{dx^2} - 4 \frac{dy}{dx} + 4y = e^{2x} + x^3 + \cos 2x$	(07)	2	3	1.

Q.3	(a)	Write down the Basic steps for solving Cauchy's Homogeneous linear differential equation.	(02)	3	1	1.1.1												
	(b)	Find the Complimentary function of $(3x + 2)^2 \frac{d^2y}{dx^2} + 3(3x + 2) \frac{dy}{dx} - 36y = 0$	(02)	3	2	1.1.1												
	(c)	Solve the Equation $L \frac{di}{dt} + Ri = E_0 \sin \omega t$, where L,R and E_0 are constants.	(03)	3	3	1.1.1												
	(d)	Solve the following Simultaneous differential equation $\frac{dx}{dt} + 5x + y = e^t, \quad \frac{dy}{dt} + x + 5y = e^{5t}$	(07)	3	2	1.1.1												
		OR																
	(e)	Apply the method of variation of parameter to solve $\frac{d^2y}{dx^2} + y = \tan x$	(07)	3	2	1.1.1												
Q.4	(a)	Find the probability of getting 4 heads in 6 tosses of a fair coin.	(02)	4	2	1.1.1												
	(b)	Derive the recurrence formula for Poisson distribution.	(02)	4	2	1.1.1												
	(c)	If 10% of bolts produced by a machine are defective. Determine the probability that out of 10 bolts, chosen at random (i) One bolts (ii) at most 2 bolts will be defective.	(03)	4	2	1.1.1												
	(d)	Derive the formula for mean and variance of Poisson Distribution.	(07)	4	2	1.1.1												
		OR																
	(e)	(i) Write any two properties of Normal distribution. (ii) By the method of least squares, Find the Straight line that best fits for the following data <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr><tr><td>y</td><td>1.0</td><td>2.9</td><td>4.8</td><td>6.7</td><td>8.6</td></tr></table>	x	0	1	2	3	4	y	1.0	2.9	4.8	6.7	8.6	(07)	4	1.3	1.1.1
x	0	1	2	3	4													
y	1.0	2.9	4.8	6.7	8.6													
Q.5	(a)	Subtract $Z_1 = \frac{3}{4} - \frac{7}{3}i$ from $Z_2 = \frac{-5}{3} + \frac{11}{5}i$	(02)	5	2	1.1.1												
	(b)	Prove that $\cosh^{-1}x = \log(x + \sqrt{x^2 - 1})$	(02)	5	2	1.1.1												
	(c)	Solve $x^4 + i = 0$	(03)	5	3	1.1.1												
	(d)	Separate $\tan^{-1}(a + ib)$ into real and imaginary parts.	(07)	5	3	1.1.1												
		OR																
	(e)	(i) Expand $\sin 5\theta$ in terms of $\sin \theta$. (ii) Sum the series $\cos \alpha + x \cos(\alpha + \beta) + \frac{x^2}{2!} \cos(\alpha + 2\beta) + \dots \infty$	(07)	5	3	1.1.1												

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FEB. 2023 EXAMINATION
I B.E./B.TECH. (4YDC) EXAM
MA 10501: MATHEMATICS - II

Exam Code 1232

Time : 3 Hrs.]

[Max. Marks : 70

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

Note : Each question carry five subparts a, b, c, d and e. Attempt subparts a, b, c and any one from d or e in each question. All questions carry equal marks.

		MARKS	CO	BL	PI
Q.1	(a) If A and B are two unitary matrices, show that AB is a unitary matrix.	02	CO1	BL-1	1.1.1
	(b) Express the matrix $A = \begin{bmatrix} 1+i & 2 & 5-5i \\ 2i & 2+i & 4+2i \\ -1+i & -4 & 7 \end{bmatrix}$ as the sum of Hermitian matrix and skew- Hermitian matrix.	02	CO1	BL-2	1.1.1
	(c) Examine the following vectors for linear dependence $X_1 = (1, 2, 4), X_2 = (2, -1, 3), X_3 = (0, 1, 2), X_4 = (-3, 7, 2)$	03	CO1	BL-2	1.1.
	(d) Discuss the consistency of the following system of equations and if consistent solve the equations : $2x - 3y + 7z = 5, 3x + y - 3z = 13, 2x + 19y - 47z = 32$	07	CO1	BL-3	1.1.
	(e) Verify Cayley-Hamilton theorem for the matrix $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ and hence find A^{-1} .	07	CO1	BL-3	1.1
Q.2	(a) Find differential equation of which $y = e^x (A \cos x + B \sin x)$ is a solution.	02	CO2	BL-1	1.
	(b) Define exact differential equation with example.	02	CO2	BL-2	1.
	(c) Solve $(1 + y^2)dx = (\tan^{-1}y - x)dy$	03	CO2	BL-2	1.
	(d) Solve $(2x \log x - xy)dy + 2yxdx = 0$	07	CO2	BL-2	1.
	(e) Solve $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = x \cos 2x$.	07	CO2	BL-2	1.
Q.3	(a) Solve $4x^2 \frac{d^2y}{dx^2} + 16x \frac{dy}{dx} + 9y = 0$	02	CO3	BL-1	1
	(b) Write the step of method of variation of parameters.	02	CO3	BL-2	1
	(c) Find the particular integral of differential equation $x^2 \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} + 2y = e^x$	03	CO3	BL-2	1

	(d)	Solve the following simultaneous differential equation $\frac{dx}{dt} + 4x + 3y = t, \frac{dy}{dt} + 2x + 5y = e^t$	07	CO3	BL-2	2.4.1
		OR				
	(e)	A constant electromotive force E volts is applied to a circuit containing a constant resistance R ohms in series and a constant inductance L henries. If the initial current is zero, show that the current builds upto half its theoretical maximum in $(L \log 2) / R$ seconds.	07	CO3	BL-2	2.4.1
Q.4	(a)	Define Normal distribution and its properties.	02	CO4	BL-1	1.1.1
	(b)	Prove that the Poisson distribution is limiting case of Binomial distribution.	02	CO4	BL-1	1.1.1
	(c)	By the method of least squares, find the straight line that best fits the following data :	03	CO4	BL-2	1.1.1
		x: 1 2 3 4 5 y: 14 27 40 55 68				
	(d)	Derive the mean and variance of Binomial Distribution.	07	CO4	BL-2	1.1.1
		OR				
	(e)	If the probability that an individual suffers a bad reaction from a certain injection is 0.001, determine the probability that out of 2000 individuals (i) Exactly 3 (ii) more than 2 individuals (iii) None (iv) more than 1 individuals will suffer a bad reaction.	07	CO4	BL-2	1.1.1
Q.5	(a)	State Demoivre's theorem and write its applications.	02	CO5	BL1	1.1.1
	(b)	Find the general value of $\log(-3)$.	02	CO5	BL-1	1.1.1
	(c)	Prove that $\tanh Z$ and $\sinh Z$ are periodic function and find their periods.	03	CO5	BL-2	1.1.1
	(d)	If $u = \log \tan\left(\frac{\pi}{4} + \frac{\theta}{2}\right)$, then prove that (i) $\tanh \frac{u}{2} = \tan \frac{\theta}{2}$ and (ii) $\cosh u = \sec \theta$.	07	CO5	BL-2	1.1.1
		OR				
	(e)	Sum the series $\sin \alpha \cos \beta - \frac{1}{2} \sin^2 \alpha \cos 2\beta + \frac{1}{3} \sin^3 \alpha \cos 3\beta - \dots \infty$.	07	CO5	BL-3	1.1.1

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NOV-DEC.2022 EXAMINATION

B.E./B.TECH.II EXAMINATION

MA-10501- MATHEMATICS-II

Time: 3 Hrs.]

Code: 510

[Max. Marks: 70]

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

S. NO.	Questions	Marks	CO	BL	PI
Q.1 (a)	Find the eigen value of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 2 \\ 0 & 0 & -2 \end{bmatrix}$	02	CO1	1	1.1.
(b)	State Cayley Hamilton theorem.	02	CO1	3	1.1.
(c)	Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$	03	CO1	3	1.1.
(d)	Investigate the value of λ and μ so that the equation : $2x + 3y + 5z = 9$ $7x + 3y - 2z = 8$ $2x + 3y + \lambda z = \mu$ have (i) no solution (ii) unique solution	07	CO1	4	2.4.
OR					
(e)	Find the eigen value and corresponding eigen vectors of the matrix $A = \begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$	07	CO1	4	2.4.
Q.2 (a)	Solve: $(D^2 - 1)y = 0$	02	CO2	1	1.1.
(b)	Check the exactness of given differential equation $(\cos x + y \sin x)dx = (\cos x)dy$	02	CO2	1	1.1.
(c)	Solve: $\frac{dy}{dx} + y \tan x = \sin 2x$	03	CO2	2	1.1.
(d)	Solve: $((D^3 - 3D^2 + 4D - 2)y = \cos x$	07	CO2	3	1.1.
OR					
(e)	Solve: $((D^2 - 6D + 9)y = 6e^{3x}$	07	CO2	3	2.1.
Q.3 (a)	Find the complementary function of differential equation $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^4$	02	CO3	1	1.1.
(b)	Write the step of variation of parameters.	02	CO3	2	1.1.

(c)	Find the particular integral of differential equation $(1+x)^2 \frac{d^2y}{dx^2} + (1+x) \frac{dy}{dx} + y = \sin 2(\log(1+x))$	03	CO3	2	1.1.1												
(d)	Apply the method of variation of parameters to solve: $\frac{d^2y}{dx^2} + y = \operatorname{cosec} x$	07	CO3	3	2.4.1												
OR																	
(e)	Solve the following simultaneous differential equation $\frac{dx}{dt} = y + 1, \quad \frac{dy}{dt} = x + 1$	07	CO3	3	2.4.1												
Q.4(a)	A die is tossed thrice. A success is getting 1 or 6 on a toss. Find the mean.	02	CO4	1	1.1.1												
(b)	Write the Recurrence formula for Binomial and Poisson distribution.	02	CO4	1	1.1.1												
(c)	If the probability of a defective bolt is 0.1 find the standard deviation for the distribution bolts in a total of 400.	03	CO4	2	1.1.1												
(d)	Fit a Poisson distribution to the following data	07	CO4	2	1.1.1												
OR																	
(e)	Find the best values of a and b so that $y = a + bx$ fits the data given in the table	07	CO4	2	1.1.1												
<table border="1"> <tr> <td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr> <tr> <td>f</td><td>122</td><td>60</td><td>15</td><td>2</td><td>1</td></tr> </table>						x	0	1	2	3	4	f	122	60	15	2	1
x	0	1	2	3	4												
f	122	60	15	2	1												
OR																	
<table border="1"> <tr> <td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td></tr> <tr> <td>f</td><td>1.0</td><td>2.9</td><td>4.8</td><td>6.7</td><td>8.6</td></tr> </table>						x	0	1	2	3	4	f	1.0	2.9	4.8	6.7	8.6
x	0	1	2	3	4												
f	1.0	2.9	4.8	6.7	8.6												
Q.5 (a)	To convert $x + iy$ into polar form.	02	CO5	1	1.1												
(b)	Prove that $\cosh x - \sinh x = e^{-x}$	02	CO5	1	1.1												
(c)	Separate into real and imaginary parts of circular function $\cos(x + iy)$	03	CO5	2	1.1												
(d)	Sum the series $\frac{\sin \theta}{L1} - \frac{\sin 2\theta}{L2} + \frac{\sin 3\theta}{L3} - \frac{\sin 4\theta}{L4} + \dots \dots \dots \infty$	07	CO5	3	1.1												
OR																	
(e)	Find the roots of complex number $x^4 + i = 0$	07	CO5	3	1.1												

Code: 937

JUNE-JULY 2022 EXAMINATION

I B.E/B. TECH EXAM

MA 10501:-MATHEMATICS-II

Time: 3 Hrs.]

[Max. Marks: 70]

TOTAL NO. OF QUESTIONS IN THIS PAPER:5

Note: Attempt all questions. All questions carry equal marks. Each question carries five subparts a, b, c, d and e. Parts a, b and c are compulsory and attempt any one from d and e.

S. No.	Questions	Marks	CO	BL	PI
Q.1	(a) Show that $A = \begin{bmatrix} -i & 3+2i & -2-i \\ -3+2i & 0 & 3-4i \\ 2-i & -3-4i & -2i \end{bmatrix}$ is skew Hermitian matrix.	(02)	CO1	1	1.1.1
	(b) Find the characteristic roots of the matrix $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$	(02)	CO1	1,2	1.1.1
	(c) State and prove Cayley Hamilton theorem.	(03)	CO1	1,2	1.1.1
	(d) Determine the values of λ and μ such that the system $2x - 5y + 2z = 8, 2x + 4y + 6z = 5, x + 2y + \lambda z = \mu$ Have: (i) No solution (ii) A Unique Solution (iii) Infinite number of solution.	(07)	CO1	3	1.1.1
	OR				
	(e) Reduce the following matrix to normal form and find its rank $A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$	(07)	CO1	4	1.1.1
Q.2	(a) Form the differential equation by eliminating the arbitrary constant $y^2 = m(a^2 - x^2)$	(02)	CO2	1	1.1.1
	(b) Solve $\frac{dy}{dx} = e^{3x-2y} + x^2 e^{-2y}$	(02)	CO2	1,2	1.1.1
	(c) Solve $[1 + \log(xy)]dx + \left[1 + \frac{x}{y}\right]dy = 0$	(03)	CO2	2	1.1.1
	(d) Solve $x \frac{dy}{dx} + y = y^2 \log x$	(07)	CO2	1,3	1.1.1
	OR				
	(e) Solve $\frac{d^2y}{dx^2} + y = e^{-x} + \cos x + x^3 + e^x \sin x$	(07)	CO2	1,3	1.1.1

Q.3	(a)	Find the Complimentary function of the differential equation $x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = 0$	(02)	CO3	1	1.1.1																				
	(b)	Write down the basic steps for solving the method of variation of parameter.	(02)	CO3	2	1.1.1																				
	(c)	Find the Particular integral of $(1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = \sin[2\{\log(1+x)\}]$	(03)	CO3	1,2	1.1.1																				
	(d)	Solve $\frac{dx}{dt} + 5x + y = e^t$, $\frac{dy}{dt} - x + 3y = e^{2t}$	(07)	CO3	3	1.1.1																				
OR																										
	(e)	A coil having a resistance of 15 ohms. And an inductance of 10 henries is connected to a 90 volts supply. Determine the value of the current after 2 seconds. ($e^{-3} = 0.05$)	(07)	CO3	1,2	1.1.1																				
Q.4	(a)	The mean and variance of a binomial distribution $P(X, n, p)$ are 4 and $\frac{4}{3}$ respectively. Find $P(X = 2)$.	(02)	CO4	1	1.1.1																				
	(b)	Derive the recurrence formula for Poisson distribution	(02)	CO4	1	1.1.1																				
	(c)	The life of army shoes is normally distributed with mean 8 months and standard deviation 2 months. If 5000 pairs are insured, how many pairs would be expected to need replacement after 12 months? (Given that $P(Z \geq 2) = 0.0228$ and $Z = \frac{x-\mu}{\sigma}$)	(03)	CO4	2	1.1.1																				
	(d)	Derive the formula of mean and variance for binomial distribution.	(07)	CO4	3	1.1.1																				
OR																										
	(e)	Fit a second degree parabola to the following data taking y as dependent variable <table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>Y</td><td>2</td><td>6</td><td>7</td><td>8</td><td>10</td><td>11</td><td>11</td><td>10</td><td>9</td></tr></table>	x	1	2	3	4	5	6	7	8	9	Y	2	6	7	8	10	11	11	10	9	(07)	CO4	3	1.1.1
x	1	2	3	4	5	6	7	8	9																	
Y	2	6	7	8	10	11	11	10	9																	
Q.5	(a)	Separate $\sec(x + iy)$ into real and imaginary parts.	(02)	CO5	2	1.1.1																				
	(b)	Express $\frac{(6+i)(2-i)}{(4+3i)(1-2i)}$ in the form of $a + ib$.	(02)	CO5	2	1.1.1																				
	(c)	Prove that $\operatorname{sech}^{-1} x = \log \frac{1+\sqrt{1-x^2}}{x}$	(03)	CO5	3	1.1.1																				
	(d)	If $\alpha, \alpha^2, \alpha^3, \alpha^4$ are the roots of $x^5 - 1 = 0$ find them and show that $(1 - \alpha)(1 - \alpha^2)(1 - \alpha^3)(1 - \alpha^4) = 5$.	(07)	CO5	3	1.1.1																				
OR																										
	(e)	Sum the series $\sin \alpha - \frac{\sin(\alpha+2\beta)}{2!} + \frac{\sin(\alpha+4\beta)}{4!} - \dots \infty$	(07)	CO5	3	1.1.1																				

ONLINE FEB. 2022 EXAMINATION

B.E. /B.TECH EXAM

MA 10501:- MATHEMATICS-II

Time:90 Minutes]

[Max. Marks: 40

PART B: Descriptive Questions

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

Note: Each question carry three subparts a, b and c. Attempt any two from a, b and c. All questions carry equal marks.

S. No.	Questions	Marks	CO	BL	P
Q.1	(a) Reduce the matrix A into its normal form where $A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 3 & 4 & 1 & 2 \\ -2 & 3 & 2 & 5 \end{bmatrix}$ and hence obtain its rank.	(04)	CO1	2	1.1
	(b) Find the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$	(04)	CO1	2,3	1.1
	(c) Investigate the values of λ and μ so that the equations $2x+3y+5z=9$; $7x+3y-2z=8$, $2x+3y+\lambda z=\mu$ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions	(04)	CO1	2	1.1
Q.2	(a) Solve $\frac{dy}{dx} = \frac{y-x+1}{y+x+5}$	(04)	CO2	2	1.1
	(b) Solve $(x^2 + y^2)dx - 2xydy = 0$	(04)	CO2	2	1.1
	(c) Solve $(D^2 + 2D + 4)y = e^x \sin 2x$	(04)	CO2	2	1.1

Q.3	(a)	Solve $x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y = x^2 + 2 \log x$	(04)	CO3	2	1.1.1																		
	(b)	Solve $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - y = x^2 e^x$ by method of variation of parameters.	(04)	CO3	3	1.1.1																		
	(c)	The equations of electromotive force in terms of current i for an electrical circuit having resistance R and a condenser of capacity C , in series is $E = Ri + \int \frac{i}{C} dt$. Find the current i at any time t , when $E = E_0 \sin wt$.	(04)	CO3	3	1.1.1																		
Q.4	(a)	Fit an exponential curve $y = ab^x$ to the following data <table border="1"><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td></tr><tr><td>y</td><td>1.0</td><td>1.2</td><td>1.8</td><td>2.5</td><td>3.6</td><td>4.7</td><td>6.6</td><td>9.1</td></tr></table>	x	1	2	3	4	5	6	7	8	y	1.0	1.2	1.8	2.5	3.6	4.7	6.6	9.1	(04)	CO4	3	1.1.1
	x	1	2	3	4	5	6	7	8															
	y	1.0	1.2	1.8	2.5	3.6	4.7	6.6	9.1															
(b)	In 800 families with 4 children each, how many families would be expected to have (i) 2 boys and 2 girls (ii) at least 1 boy (iii) no girl. Assuming equal probabilities for girls and boys.	(04)	CO4	2	1.1.1																			
	(c)	If 2% of light bulbs are defective, find the probability that (i) at least one is defective and (ii) exactly 7 are defective. Also find $P(1 < X < 8)$ in a sample of 100.	(04)	CO4	2	1.1.1																		
Q.5	(a)	Express $\text{Log}(\text{Log} i)$ in the form $A + iB$.	(04)	CO5	2	1.1.1																		
	(b)	Sum the series $1 + \frac{\cos \alpha}{\cos \alpha} + \frac{\cos 2\alpha}{2! \cos^2 \alpha} + \frac{\cos 3\alpha}{3! \cos^3 \alpha} + \dots - \infty$	(04)	CO5	2	1.1.1																		
	(c)	If $u = \log_e \left[\tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \right]$ prove that $\tanh \frac{u}{2} = \tan \frac{\theta}{2}$	(04)	CO5	2	1.1.1																		

ONLINE JULY 2021 EXAMINATION

B.E/B.TECH EXAM

MA 10501:- Mathematics-II

Time : 90 Minutes]

[Max. Marks

PART B: Descriptive Questions

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

NOTE:- Each question carry three subparts a, b and c. Attempt any two from a, b and c. All questions carry equal marks.

S. NO.

Questions

- Q.1 (a) Find the rank of the matrix by reducing it to normal form

$$A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 3 & 4 & 0 & -1 \\ -1 & 0 & -2 & 7 \end{bmatrix}$$

- (b) Show that the following system of equation is inconsistent.

$$2x - y + z = 9, \quad 3x - y + z = 6, \quad 4x - y + 2z = 7, \quad -x + y - z = 4$$

- (c) Find the Eigen value and corresponding Eigen vector of the matrix:

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

- Q.2 (a) Solve: $(x^2y^2 + 3y)dx + (3x^2y - x)dy = 0$

- (b) Solve: $(x^2 + 2x + 1)y = x^2 \cos x$

- (c) Solve: $\frac{dy}{dx} - \frac{y}{x} + 2y = \cos(x^2 - y^2)$

- Q.3 (a) Solve: $(1 + 2x) \frac{dy}{dx} = y(1 + 2x)^2$

Solve : $\frac{dx}{dt} + 5x + y = e^t$, $\frac{dy}{dt} - x + 3y = e^{2t}$ (04)

- (c) A circuit consisting of a resistance R and a condenser of capacity c is connected in series with a voltage E. Assuming that there is no charge on the condenser at $t=0$ find the value of the current I, voltage and charge q at any time t. (04)

- 2.4(a) The probability of a man hitting a target is $\frac{1}{3}$ (i) If he fires times, What is the probability of his hitting the target at least twice (ii) How many times must he fire so that the probability of his hitting the target at least once is more than 90%. (04)

- (b) The following data are the number of seeds germinating out by 10 on damp filter for 80 sets of seeds. Fit a binomial distribution of these data. (04)

x:	0	1	2	3	4	5	6	7	8	9	10
y:	6	20	28	12	8	6	0	0	0	0	0

- (c) By using the method of least square to fit a second degree parabola for the following data (04)

x:	0	1	2	3	4	5	6	7	8
y:	12.0	10.5	10.0	8.0	7.0	8.0	7.5	8.5	9.5

- Q.5(a) Prove that $\frac{1+\cos 6\theta}{1+\cos 2\theta} = 16\cos^4\theta - 24\cos^2\theta + 9$ (04)

- (b) Show that $x^5 - 1 = (x-1)[x^2 + 2x\cos\frac{\pi}{5} + 1][x^2 + 2x\cos\frac{3\pi}{5} + 1]$ (04)

- (c) Find the sum of the series $\cos\alpha + \frac{1}{2}\cos 3\alpha + \frac{1}{2^2}\cos 5\alpha + \dots \dots \dots \infty$. (04)

Time : 3 Hrs.]

NOV. 2020 EXAMINATION

B.E./B.Tech.IV EXAMINATION

MA 10501:- MATHEMATICS-II

Code-832

[Max. Marks : 70]

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

S. NO.	Questions	Marks	CO	BL
Q.1 (a)	Define idempotent matrix and Prove that the eigen value of an idempotent matrix are either zero or unity	02	CO1	BL-1,2
(b)	Find the rank of the matrix $A = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 9 \\ -1 & -3 & -4 & -3 \end{bmatrix}$	02	CO1	BL-2
(c)	Solve by matrix method $\begin{aligned} 2x + 3y + z &= 9 \\ x + 2y + 3z &= 6 \\ 3x + y + 2z &= 8 \end{aligned}$	03	CO1	BL-2,3
(d)	Find the value of k for which the following system of equation as consistent and has non-trivial solution solve equation for all such value of k $\begin{aligned} (k-1)x + (3k+1)y + 2kz &= 0 \\ (k-1)x + (4k-2)y + (k+3)z &= 0 \\ 2x + (3k+1)y + 3(k-1)z &= 0 \end{aligned}$	07	CO1	BL-2,3
OR				
(e)	Find two non-singular matrix P and Q such that PAQ is in the normal form where $A = \begin{bmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}$ And Hence find its rank	07	CO1	BL-2,3

5 = 2
6 = 14 - 1

Q.2 (a)	Solve: $y - x \frac{dy}{dx} = a(y^2 + \frac{dy}{dx})$	02	CO2	BL	
(b)	Solve: $(x + y + 1) \frac{dy}{dx} = 1$	02	CO2	BL-2	
(c)	Solve: $(3x + 2y^2)ydx + 2x(2x + 3y^2)dy = 0$	03	CO2	BL-2,3	1.1
(d)	Solve: $\frac{d^2y}{dx^2} - 6 \frac{dy}{dx} + 13y = 8e^{3x} \sin 4x + 2^x$	07	CO2	BL-2,3	1.1
OR					
(e)	Solve: $(D^2 - 4D + 3)y = 2xe^x + 3e^x \sin x$	07	CO2	BL-2,3	1.1
Q.3 (a)	Solve: $(x^2 D^2 + 2xD)y = \log x$	02	CO3	BL-2	1.1
(b)	Define Application of differential equation in electrical circuit.	02	CO3	BL-1	1.1
(c)	Solve $(D^2 + 1)y = \operatorname{cosec} x$ using method of variation parameter..	03	CO3	BL-2	1.1
(d)	Solve: $\frac{d^2x}{dt^2} - 3x - 4y = 0$ $\frac{d^2y}{dt^2} + x + y = 0$	07	CO3	BL-2	1.1
OR					
(e)	Solve: $x^2 \frac{d^2y}{dx^2} + 2x \frac{dy}{dx} = xe^x$	07	CO3	BL-3	1.1
Q.4(a)	Define Binomial Distribution with example and write the condition for binomial distribution.	02	CO4	BL-1	1.1
(b)	Write properties of Normal distribution and define the mean and variance of Poisson distribution.	02	CO4	BL-1	1.1
(c)	If 2% of light bulbs are defective find the probability that (i) at least one is defective (ii) exactly 7 are defective.	03	CO4	BL-3	1.1
(d)	The mean yield for a one acre plot is 662 kg with an S.D of 32kg Assuming normal distribution how many one acre plots in a batch of 1000 plots would you expect to have yield (i) over 700kg and (ii) below 650kg (iii) what is the lowest yield of the best 100 plots	07	CO4	BL2, -3	1.1

BL-2

OR

Fit a curve of the form $y = ae^{bx}$ to the following data

x	1	3	5	7	9
y	115	105	95	85	80

07

CO4

BL2,
-3

15 (a)

Expand $\frac{\sin 5\theta}{\sin \theta}$ in power of $\cos \theta$ only.

02

CO5

BL-1

(b)

Solve the following equation

$$x^6 - i = 0$$

02

CO5

BL-2

(c)

If $\alpha, \alpha^2, \alpha^3, \alpha^4$ are roots of $x^5 - 1 = 0$ then show that
 $(1-\alpha)(1-\alpha^2)(1-\alpha^3)(1-\alpha^4) = 5$

03

CO5

BL-
2,3

(d)

Find all the roots of $x^{12} - 1 = 0$ and identify the roots which are also the roots of $x^4 - x^2 + 1 = 0$

07

CO5

BL-2

OR

(e)

Find the sum of the series $a \cos^2 \alpha - \frac{1}{3} a^3 \cos^2 3\alpha + \frac{1}{5} a^5 \cos^2 5\alpha$
 $+ \dots - \infty$.

07

CO5

BL-3

Time: 3 Hrs]

[Max Marks: 70

TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

Note: Attempt all the questions. Each question carries four subparts. Attempt anyone from 'a' and 'b' and similarly anyone from 'c' and 'd'.

- Q.1. (a) Find the inverse of a matrix M by applying elementary transformation (7)

$$M = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 1 & 1 & -1 & -2 \\ 1 & 2 & 0 & 1 \\ -1 & 1 & 2 & 6 \end{bmatrix}$$

OR

- (b) Find non-singular matrices P and Q so that PAQ is a normal form, and hence find its rank (7)

$$A = \begin{bmatrix} 2 & 1 & -3 & -6 \\ 3 & -3 & 1 & 2 \\ 1 & 1 & 1 & 2 \end{bmatrix}$$

- (c) Find all the Eigen values and Eigen vectors of the following matrix: (7)

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}$$

OR

- (d) Verify Cayley Hamilton theorem (7)

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 2 \end{bmatrix}$$

- Q.2. (a) Solve $x \cos \frac{y}{x} (y dx + x dy) = y \sin \frac{y}{x} (x dy - y dx)$ (7)

OR

- (b) Find the value of λ for the given differential equation. Solve the equation for this value of λ : $(xy^2 + \lambda x^2 y) dx + (x + y)x^2 dy = 0$ (7)

- (c) Solve $x \frac{dy}{dx} + y = y^2 \log x$ (7)

OR

- (d) Solve $\frac{d^3 y}{dx^3} - 3 \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - y = (1+x)e^x$ (7)

Q.3. (a) Solve $x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 12y = x^3 \log x$ (7)

OR

(b) Solve the following simultaneous differential equations (7)

$$\frac{dx}{dt} + 5x + y = e^t; \quad \frac{dy}{dt} + x + 5y = e^{3t}$$

(c) Apply method of variation of parameter to solve (7)

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = \frac{e^x}{1+e^x}$$

OR

(d) Find the current at any time $t > 0$ in a circuit having in series a constant electromotive force 40V, a resistor 10Ω , and an inductor $0.2H$, given that the initial current is zero. Find the current when $E(t) = 150 \cos 200t$ (7)

Q.4. (a) Six dice are thrown 729 times. How many times do you expect at least three dice to show a five or six? (7)

OR

(b) The Poisson distribution of number of road accidents per day in a city with mean (7)

4. Find the number of days out of 100 days when there will be: ($e^{-4} = 0.018$)

(i) no accident

(ii) at least 2 accidents

(iii) at most 3 accidents

(iv) exactly 4 accidents

(c) Define Normal distribution. Draw the normal curve and write the properties of normal curve. (7)

OR

(d) Fit a second degree parabola to the following data: (7)

x	0	1	2	3	4
y	-4	-1	4	11	20

Q.5. (a) If $\alpha, \alpha^2, \alpha^3, \alpha^4$ are the roots of $x^5 - 1 = 0$, then show that (7)

$$(1-\alpha)(1-\alpha^2)(1-\alpha^3)(1-\alpha^4) = 5$$

OR

(b) If $e^z = \sin(u+iv)$ and $z = x+iy$, then prove that $2e^{2x} = \cosh 2v - \cos 2u$. (7)

(c) Separate $\tan^{-1}(a+ib)$ into real and imaginary parts. (7)

OR

(d) Sum the series (7)

$$x \sin \theta - \frac{1}{2} x^2 \sin 2\theta + \frac{1}{3} x^3 \sin 3\theta - \dots \infty$$

APRIL 2019 EXAMINATION
I.B.E. (4YJC) EXAM
MA10501 : MATHEMATICS-II

Code 607

Time : 3 Hrs.]

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

[Max. Marks : 70

Note : Attempt all questions. All questions carry equal marks. Each question carry four subparts a, b, c and d. Attempt any one from a and b and similarly any one from c and d.

- Q.1 (a) Reduce the following matrix into its normal form and hence find its rank (07)

$$A = \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

OR

- (b) Determine the values of α and β for which the system

$$\begin{bmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & \alpha \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \beta \\ 3 \\ -1 \end{bmatrix}$$

has (i) a unique solution (ii) no solution (iii) infinitely many solutions.

- (c) (i) Show that every square matrix can be expressed as the sum of a Hermitian matrix and a skew Hermitian matrix in one and only one way. (07)

(ii) Let A be a 3×3 matrix such that $\det(2A - I_3) = 0$. If $\text{trace}(A) = 15$ and $\det(A) = 105$, then find the eigen values of A .

OR

- (d) State Cayley Hamilton theorem and using it find A^{-1} , where

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix} \text{ is a square matrix.}$$

- Q.2 (a) Solve (07)

$$\frac{dy}{dx} + \frac{1}{x} \tan y = \frac{1}{x^2} \tan y \sin y$$

OR

- (b) Solve

$$\frac{dy}{dx} = \frac{x^2 + xy}{x^2 + y^2}$$

- (c) Solve (07)

$$(3x + 2y^2)ydx + 2x(2x + 3y^2)dy = 0$$

OR

- (d) Solve

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = x.e^x \cdot \sin x$$

Q.3 (a) Solve

$$x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 12y = x^3 \log x$$

OR

(b) Solve by method of variation of parameters

$$\frac{d^2 y}{dx^2} - y = \left(1 + \frac{1}{e^x}\right)^{-2}$$

(c) A mechanical system with two degree of freedom satisfies the equation

$$2 \frac{d^2 x}{dt^2} + 3 \frac{dy}{dt} = 4 \text{ and } 2 \frac{d^2 y}{dt^2} - 3 \frac{dx}{dt} = 0.$$

Obtain expressions for x and y in terms of t , given $x, y, \frac{dx}{dt}, \frac{dy}{dt}$ all vanish at $t = 0$.

OR

(d) A moving body is opposed by a force per unit mass of values cx and resistance per unit mass of values bv^2 when x and v are the displacement and velocity of the particle at that instant. Find the velocity of the particle in terms of x , if it starts from rest.

Q.4 (a) If the mean and variance of a binomial distribution are 4 and 2 respectively, find the probability of (i) exactly 2 successes (ii) less than 2 successes (iii) at least 2 successes.

OR

(b) An insurance company found that only 0.01% of the population is involved in a certain type of accident each year. If its 1000 policy holders were randomly selected from the population, what is the probability that not more than two of its clients are involved in such an accident next year? (given that $e^{-0.1} = 0.9048$)

(c) Fit a second degree parabola to the following data:

x :	10	15	20	25	30	35	40
y :	11	13	16	20	27	34	41

OR

(d) A manufacturer of envelopes knows that the weight of the envelopes is normally distributed with mean 1.9 gm and variance 0.01gm. Find how many envelopes weighing (i) 2 gm or more (ii) 2.1 gm or more, can be expected in a given packet of 1000 envelopes. [Given : If t is the normal variable, then $\phi(0 \leq t \leq 1) = 0.3413$ and $\phi(0 \leq t \leq 2) = 0.4772$].

Q.5 (a) Solve $x^7 = 1$ and prove that the sum of n th powers of the roots is 7 or zero, according as n is or is not multiple of 7.

OR

(b) If $\cosh x = \sec \theta$, prove that

$$(i) \theta = \frac{\pi}{2} - 2 \tan^{-1}(e^{-x}) \quad (ii) \tanh \frac{\pi}{2} = \tan \frac{\theta}{2}$$

(c) Express $\text{Log}(\text{Log } i)$ in the form of $A + iB$.

OR

(d) Sum the series

$$\cos \theta + \sin \theta \cos 2\theta + \frac{\sin^2 \theta \cos 3\theta}{1.2} + \dots \infty$$

NOV - DEC 2018 EXAMINATION
1 Year B.E./B.Tech. (Common for all Branches)
MA10501: MATHEMATICS - II

1086

Time: 3 Hrs]

TOTAL NO. OF QUESTIONS IN THIS PAPER: 5

[Max Marks: 70

Note: Attempt all the questions. Each question carries four subparts. Attempt any one from 'a' and 'b' and similarly anyone from 'c' and 'd'.

- Q.1. (a) Find the normal form of the following matrix and hence find its rank: (7)
- $$A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix}$$

OR

- (b) Show that the vectors $x_1 = (2, 3, 1, -1)$, $x_2 = (2, 3, 1, -2)$, $x_3 = (4, 6, 2, 1)$ are linearly dependent and row vectors of the following matrix are linearly independent. (7)

$$A = \begin{bmatrix} 2 & -6 & -2 & -3 \\ 5 & -13 & -4 & -7 \\ -1 & 4 & 1 & 2 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

- (c) Find all the Eigen values and Eigen vectors of the following matrix: (7)

$$A = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 3 \end{bmatrix}$$

OR

- (d) Show that the Cayley-Hamilton theorem is satisfied by the given matrix. Hence find A^{-1} (7)

$$A = \begin{bmatrix} 1 & 3 & 7 \\ 4 & 2 & 3 \\ 1 & 2 & 1 \end{bmatrix}$$

- Q.2. (a) Solve $\frac{dy}{dx} = \frac{x^2 + xy}{x^2 + y^2}$ (7)

OR

- (b) Solve $\frac{dy}{dx} = \frac{y+1}{(y+2)e^y - x}$ (7)

- (c) Solve: $(3x^2y^4 + 2xy)dx + (2x^3y^3 - x^2)dy = 0$ (7)

OR

- (d) A body executes damped forced vibrations given by the equation (7)

$$\frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + b^2x = e^{-kt} \sin \omega t$$

Solve the differential equation for both the cases when $\omega^2 = b^2 - k^2$ and $\omega^2 \neq b^2 - k^2$.

Q.3. (a) Solve: $(x+1)^2 \frac{d^2y}{dx^2} + (x+1) \frac{dy}{dx} = (2x+3)(2x+4)$ (7)

OR

(b) Solve the following simultaneous differential equations: (7)

$$\frac{d^2x}{dt^2} + \frac{dy}{dt} + 3x = e^{-t}; \quad \frac{d^2y}{dt^2} - 4 \frac{dx}{dt} + 3y = \sin 2t$$

(c) Solve the following differential equation by using method of separation of variables:

$$x^2 y'' + xy' - y = x^3 e^x$$

OR

(d) A condenser of capacity C farads with V_0 is discharged through a resistance R ohms. Show that if q coulomb is the charge on the condenser, i ampere the current and V the voltage at time t. (7)

$$q = CV, V = Ri \text{ and } i = \frac{dq}{dt}, \text{ hence show that } V = V_0 e^{-\frac{t}{RC}}$$

Q.4. (a) If the probability that an individual suffers a bad reaction from a certain injection is 0.001, determine the probability that out of 2000 individuals (a) exactly 3; (b) more than 1; (c) more than 2; individuals will suffer a bad reaction. ($e^{-2} = 0.135$) (7)

OR

(b) Derive the expression to find the mean and variance of Binomial Distribution. (7)

(c) The mean yield per plot of a crop is 17kg and standard deviation is 3kg. if the distribution yield per plot is normal, find the percentage of plots giving yields: (7)
(i) between 15.5kg and 20kg (ii) more than 20kg. (Where $A_{0.5} = 0.1915$, $A_{1.0} = 0.3413$)

OR

(d) Fit a second degree parabola to the following data: (7)

x	10	15	20	25	30	35	40
y	11	13	16	20	27	34	41

Q.5. (a) By using De-Moivre's theorem solve the equation $x^7 - 1 = 0$. (7)

OR

(b) If $\sin(\theta + i\phi) = \cos \alpha + i \sin \alpha$, (7)
then prove that $\cos^2 \theta = \sin^2 \alpha = \sinh^2 \phi$.

(c) Separate $\operatorname{sech}(x + iy)$ into real and imaginary parts. (7)

OR

(d) Sum the series (7)

$$c \sin \theta - \frac{1}{2} c^2 \sin 2\theta + \frac{1}{3} c^3 \sin 3\theta - \dots \infty$$

ELITE.

APRIL 2018 EXAMINATION
I B.E. (4YDC) EXAM
MA10501 : MATHEMATICS-II

Code 1371

Time : 3 Hrs.]

[Max. Marks : 70]

TOTAL NO. OF QUESTIONS IN THIS PAPER : 5

Note : Attempt all questions. All questions carry equal marks. Each question carry four subparts a, b, c and d. Attempt any one from a and b and similarly any one from c and d.

- Q.1 (a) Use Gauss Jordan reduction method to compute the inverse of the matrix
- $$\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$
- (07)

- (b) Examine the following vectors for linear dependence and find the relation if it exists
 $X_1 = (1, 2, 4)$, $X_2 = (2, -1, 3)$, $X_3 = (0, 1, 2)$, $X_4 = (-3, 7, 2)$

- (c) Express the matrix
- $$A = \begin{bmatrix} 1+i & 2 & 5-5i \\ 2i & 2+i & 4+2i \\ -1+i & -4 & 7 \end{bmatrix}$$
- as the sum of Hermitian matrix and skew-Hermitian matrix.
- (07)

- (d) Solve the system of equations
 $3x+3y+2z=1$, $x+2y=4$, $10y+3z=-2$, $2x-3y-z=5$.

- Q.2 (a) Solve $xdy - ydx = \sqrt{x^2 + y^2} dx$
- (07)

- (b) Solve
- $$(1-x^2) \frac{dy}{dx} + xy = xy^2$$

- (c) Solve $(x^2y^2 + xy + 1)ydx + (x^2y^2 - xy + 1)x dy = 0$
- (07)

- (d) Solve $(D^2 - 6D + 13)y = 8e^{3x} \sin 4x$

- Q.3 (a) Solve
- $$(1+x)^2 \frac{d^2y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \log(1+x)$$
- (07)

- (b) Solve by method of variation of parameters
 $(D^2 + 2D + 1)y = e^{-x} \log x$

- (c) Solve the following simultaneous differential equation
- $$Dx + Dy + 3x = \sin t$$
- $$Dx + y - x = \cos t$$
- (07)

OR

- (d) A constant electromotive force E volts is applied to a circuit containing a constant resistance R ohms in series and a constant inductance L henries. If the initial current is zero, show that the current builds up to half its theoretical maximum in $(L \log 2) / R$ seconds.

Q.4 (a) Prove that the Poisson distribution is limiting case of Binomial distribution.

OR

- (b) Derive the normal equations for the straight line and fit a straight line to the following data

X:	1	2	3	4	5	6
Y:	6	4	3	5	4	2

- (c) Out of 800 families with 4 children each, how many families would be expected to have (i) 2 boys and 2 girls (ii) at least one boy (iii) no girl (iv) at most two girls? Assume equal probabilities for boys and girls

OR

- (d) A manufacturer of envelopes knows that the weight of the envelopes normally distributed with mean 1.9 gms and variance 0.01 gm. Find how many envelopes weighing i) 2 gm or more ii) 2.1 gm or more, can be expected in a given packet of 1000 envelopes.

[Given : if t is the normal variable then $\phi(0 \leq t \leq 1) = 0.3413$ and $\phi(0 \leq t \leq 2) = 0.4772$]

Q.5 (a) Show that $\log_e \omega$ is a many valued function. Express $\text{Log}(\text{Log} i)$ in the form of $A + iB$.

OR

- (b) State and prove De Moivre's Theorem. If n is positive integer, prove that

$$(\sqrt{3} + i)^n + (\sqrt{3} - i)^n = 2^{n+1} \cos \frac{n\pi}{6}$$

- (c) If $u = \log \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$, then prove that (i) $\tanh \frac{u}{2} = \tan \frac{\theta}{2}$ (ii) $\cosh u = \sec \theta$

OR

- (d) Sum the series $\sin \alpha \cos \beta - \frac{1}{2} \sin^2 \alpha \cos 2\beta + \frac{1}{3} \sin^3 \alpha \cos 3\beta - \dots \infty$.
